

USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
MIDTERM PROJECT (assigned Lesson 9) - Antenna Pattern Measurement
Project Description
Fall 2026

Due 2 October, 2359. Individual. Worth 30% of the course grade and 40% of your midterm grade, scored on the **100-point rubric in Section 12**. There is no resubmission – the grade you earn is the grade.

1 The assignment in one paragraph

Build a 915 MHz wire dipole. Trim it until it resonates, and prove it did. Then choose a second frequency, leave the antenna exactly as it is, and build a matching network that presents $50\ \Omega$ at that new frequency instead. Model the antenna in NEC at both frequencies. Take it into the chamber and measure its pattern at both frequencies, co-polarized and cross-polarized, against a reference antenna of known gain. Then write up what you predicted, what you measured, and what you make of the difference.

You are not designing an antenna. Lesson 7 gave you the design and Lesson 8 let you model it, so the pattern, the gain, and the impedance are all predictable before you measure anything. The work of this project is the comparison.

2 The engineering question

There are two ways to make an antenna present $50\ \Omega$ at the connector, and this project makes you do both to the same piece of wire.

At 915 MHz you change the *antenna*: you cut it until its own reactance vanishes. At your second frequency f_2 you change the *network*: the antenna stays wrong and a pair of components hides it from the source. Both roads end at $VSWR \leq 2$ looking into the connector. They do not end at the same place anywhere else.

Your report answers this question: what did each method cost you, and what did the radiated pattern do about it? A VSWR reading at the connector is a statement about power accepted, not about power radiated or where it went. Your pattern measurements are how you tell the difference.

That question is the spine of the report. Everything below exists to give you the evidence to answer it.

3 What this project assesses

The project is where the Module 2 measurement objectives are demonstrated end to end:

- I can set up a one-port sweep, run an SOL calibration at the correct reference plane, and verify the calibration before I trust a reading.
- I can read resonance, impedance, and $VSWR \leq 2$ bandwidth off a trace and off the Smith chart, and say from the sign of the reactance which way to trim.
- I can verify a range geometry is valid for both antennas before I take data, using all three far-field criteria.
- I can acquire principal-plane cuts and normalize, plot, and annotate them in dB down from the peak.
- I can extract half-power beamwidth, first sidelobe level, front-to-back ratio, gain by comparison, and cross-polarization discrimination from measured data.
- I can measure my own noise floor, state the resulting dynamic range, and say which of my numbers clears it and by how much.

4 Phase 1 - Build the dipole

Predict before you cut. Lesson 7 gives the half-wave dipole's dimensions, its $73 + j42.5 \Omega$ input impedance, and its 2.15 dBi directivity. Work out the numbers for 915 MHz on paper first.

Build it from wire and an SMA connector. The example antenna on the bench shows the feed construction: arms soldered to the center pin and to the body, a clean symmetric junction, arms straight and colinear. Make yours mechanically rigid – an arm that moves when you move the coax makes every later measurement a different antenna.

Required: the predicted arm length and predicted Z_{in} at 915 MHz, computed *before* the first cut, quoted in the report next to what you actually built.

Cut long. You can remove wire; you cannot put it back.

5 Phase 2 - Find and fix the resonance

Use the handheld VNA for this phase. It lives at the bench, it sweeps fast, and trimming is an iterative loop you will run several times.

1. Run a short-open-load calibration and verify it. State your reference plane – the numbers mean nothing without it, and the length of cable past that plane is pure rotation on the chart.
2. Sweep the antenna. Find the frequency where the reactance crosses zero. That is the resonance, and it is almost certainly not 915 MHz on the first try.
3. Read the sign of the reactance at 915 MHz and let it tell you which way to trim.
4. Trim, re-measure, repeat.

Log every iteration. Arm length, resonant frequency, impedance at resonance, VSWR at 915 MHz. You need four or five of these rows to make the trim history plot, and the plot is worth more than the final number: it shows a physical trend you can compare against the $\lambda/2$ prediction.

The bar: $\text{VSWR} \leq 2$ at 915 MHz, measured, with the trace shown. Switch the analyzer to VSWR format at the bench so the limit line is simply 2.

Note what the textbook antenna does here. An ideal $\lambda/2$ dipole at $73 + j42.5 \Omega$ on a 50Ω system is VSWR 2.18 – it *fails* this bar. Trimming to resonance is what passes it. Say in your report what your trimmed antenna's resistance came out to be and whether that is consistent with the 73Ω Lesson 7 derived.

6 Phase 3 - Build a balun

A coaxial cable is unbalanced and a dipole is balanced. Connect them directly and current that should have stayed on the inside of the shield flows back down its outside, which radiates. The thing you then measure in the chamber is not your dipole. It is your dipole plus a length of coax, and it will show up as a pattern that is tilted, asymmetric, and dependent on how the cable is routed that day.

You will build a balun. Two constructions work at 915 MHz; pick one.

Quarter-wave sleeve balun. A conducting sleeve (copper tape over a tube, or thin brass tubing) coaxial with the feed line, shorted to the shield at the end away from the antenna and open at the feed end. Length is a quarter wavelength in air at 915 MHz – compute it. The shorted quarter wave presents a high impedance at the feed, which is what chokes the outside current.

Coiled-coax choke. Wind several turns of the feed coax into a tight solenoid a few centimeters across, close to

the feed point. The coil's inductance and its self-capacitance form a high impedance to common-mode current. It is quicker to build than a sleeve and its performance is less predictable, which is why the verification below matters.

Required evidence that it works. Measure the antenna's VSWR at 915 MHz while you move, coil, and grip the feed line, with and without the balun. Report how much the reading swings in each case. A working balun makes the antenna's behavior stop depending on its cable. If you have chamber time for it, the stronger evidence is a pattern cut taken both ways.

Do not skip this and do not do it last. The balun is on the antenna for every measurement that follows, including the trim verification you may need to repeat once it is fitted.

7 Phase 4 – Detune, predict, and match

Now the second half of the engineering question.

7.1 Choose f_2

Pick any f_2 between 800 and 1050 MHz that is at least 5% away from 915 MHz – that is, outside 869–961 MHz. Different cadets should pick different frequencies. Declare your choice at the Lesson 14 checkpoint.

7.2 Predict $Z_a(f_2)$ before you design anything

Model your antenna – the one you actually built, at the length you actually trimmed it to – in the course NEC simulator and read its input impedance at f_2 . This prediction is the load your matching network is designed against.

This ordering is not a formality. A matching network designed against a measured impedance is a fitting exercise; one designed against a prediction and then measured is an experiment. Do it in that order and report both numbers.

7.3 Design, build, verify

Use the L-network method from the Lesson 4 lab. Your network must absorb the antenna's own reactance at f_2 as well as transform its resistance, so work from the full complex $Z_a(f_2)$, not from its real part. Build it on the SMA proto board with discrete components; choose the nearest stocked values and report the difference between the values you designed and the values you soldered.

Two elements should reach any f_2 in the allowed window. If yours will not, say why in the report before adding a third.

Verify on the Copper Mountain VNA. This is your final, reportable impedance data, so calibrate carefully and state the reference plane again.

The bar: $\text{VSWR} \leq 2$ at f_2 , measured on the Copper Mountain, with the trace shown and the $\text{VSWR} \leq 2$ bandwidth quoted. Compare that bandwidth against the resonant antenna's bandwidth at 915 MHz. They will not be the same, and the difference is one of the costs the engineering question is asking about.

7.4 Measure the same antenna on both instruments

Once the antenna is final, sweep it on *both* the handheld and the Copper Mountain, calibrated to the same reference plane, over the same span. Overlay the two traces.

They will not lie on top of each other. Report the disagreement – where it is worst, how large it is, and what you think accounts for it. Two instruments disagreeing about one unchanged object is the most honest uncertainty estimate you will get in this course, and it is worth more than any figure you could quote from a data sheet.

8 Phase 5 – Simulate

Use the course NEC2 simulator. Model the antenna you built: your trimmed length, your wire diameter, your feed geometry.

Produce, at both 915 MHz and f_2 :

- input impedance and VSWR referenced to 50 Ω ,
- E-plane and H-plane pattern cuts,
- directivity, and the gain if you model loss.

State your model's assumptions where they differ from the hardware: perfect conductor, no balun, no connector, no feed cable, free space rather than a chamber. Every one of those is a candidate explanation for a later disagreement, and a disagreement you predicted is worth far more than one you discovered.

9 Phase 6 – Measure the pattern

9.1 Before you take data

Verify the range geometry using all three far-field criteria from Lesson 9, for *both* the antenna under test and the reference antenna, at *both* frequencies. State which criterion governs. A dipole is small compared to a wavelength, so $2D^2/\lambda$ is probably not the one that binds – find out which is.

9.2 Measure your floor

Before the pattern, measure the floor: with the source off, or with the antennas cross-polarized and rotated away, record the level the receiver reports. Quote it in dB below your co-polarized peak. That number is your dynamic range, and it bounds every quantity you extract afterward.

Draw the floor as a horizontal line on every pattern plot in your report. A pattern measurement has a floor; the chamber contributes one and the receiver contributes the other, and whichever is higher is the one you have. Any number that lands within a few dB of it is a *bound*, not a value, and must be quoted as " $\leq$ " in your tables.

9.3 The cuts

At both 915 MHz and f_2 :

- E-plane cut, co-polarized,
- H-plane cut, co-polarized,
- both planes again, cross-polarized.

Keep the range, the cabling, and the source power identical between the co- and cross-polarized runs. If you change anything, you have changed the reference the two are compared against and the discrimination number is meaningless.

9.4 The reference antenna

Measure the reference antenna's own pattern as well. It is a five-minute swap and a five-minute sweep, and it buys you two things: a check that your range is behaving, since you know what that antenna's pattern should look like, and the received-power reference for gain by comparison.

Gain by substitution: replace your antenna with the reference, change nothing else, and record the received power. Then

$$G_{\text{AUT}} [\text{dBi}] = G_{\text{ref}} [\text{dBi}] + (P_{\text{AUT}} - P_{\text{ref}}) [\text{dB}].$$

Note the conditions this is valid under — Lesson 9 lists them — and say whether yours were met.

10 Phase 7 - Analysis

This is the part of the report that carries the grade.

10.1 Normalize honestly

The chamber gives you relative received power. NEC gives you absolute directivity. They are not the same quantity and you cannot simply plot them on one axis.

State your normalization basis in the caption of every overlay. The default: normalize the measured co-polarized cut to its own peak, normalize the simulated cut to its own peak, and plot both in dB down from peak. That comparison is about pattern *shape* and it is honest. Absolute level is a separate claim, made separately, from the gain substitution measurement.

What you must not do is slide one curve against the other until they agree. Shape agreement and gain agreement are two different findings, and collapsing them into one plot hides whichever of them failed.

10.2 Extract the numbers

From the measured cuts, at both frequencies: half-power beamwidth in both planes, null depths and positions, first sidelobe level in dBc, front-to-back ratio, cross-polarization discrimination, and gain. For each, say whether it clears your floor.

10.3 Reconcile

Put prediction and measurement side by side and account for the differences. Lesson 7's closed form, your NEC model, and your chamber data are three independent statements about one antenna. Where they disagree, work out which one you believe and why.

Candidate explanations you should be able to rule in or out: the feed cable and balun, chamber reflections and the quiet-zone limits, the range geometry, alignment and polarization error, the model's idealizations, connector and network loss at f_2 , and your own repeatability.

This is where the grade is won. You are not required to make the numbers agree. You are required to know why they do not. A disagreement you have diagnosed is a result; a disagreement you have not mentioned is a missing result.

There is no error budget to fill in. The errors are there whether or not you tabulate them – what the report needs is your reasoning about which of them is responsible for what you actually see.

11 The report

One typed technical report, submitted as a PDF, plus your data files. There is no page limit and no template: the structure is yours to choose, and choosing it is part of the work. Two pages will not do it.

Nothing handwritten. Every figure is generated by software. Every axis is labeled with units. Every trace is in a legend. Every figure and table is numbered and has a caption that says what the reader should take from it. An uncaptioned instrument screenshot is not a figure.

11.1 Required figures

1. Annotated photograph of the antenna as built, with dimensions and the balun visible.
2. Trim history: VSWR against frequency for each cut, on one set of axes, with the $VSWR = 2$ line drawn.
3. Smith chart at f_2 : the antenna unmatched and the antenna matched.
4. Matching network schematic, with designed and as-built component values.
5. Handheld and Copper Mountain traces of the final antenna, overlaid, same reference plane.
6. The NEC model geometry.
7. Simulated E- and H-plane patterns at both frequencies.
8. Measured co- and cross-polarized cuts, both planes, both frequencies, with the measured floor drawn on each.
9. Measured against simulated, overlaid, normalization basis stated in the caption.
10. The reference antenna's measured pattern.

11.2 Required tables

1. **Predicted against measured.** Resonant frequency, input impedance, VSWR, $VSWR \leq 2$ bandwidth, half-power beamwidth in both planes, first sidelobe level, front-to-back ratio, cross-polarization discrimination, and gain – with three columns: Lesson 7 closed form, NEC, and measured. Both frequencies.
2. **The match at f_2 .** Predicted $Z_a(f_2)$, designed component values, as-built component values, achieved VSWR, achieved bandwidth.

11.3 Data files

Submit alongside the PDF, named so a reader can tell what each one is: Touchstone .s1p files from both analyzers, your pattern data as CSV, and your NEC input decks. Your data files must be your own measurements.

12 Grading

The project is scored out of 100 on the rubric below. Each row is worth the points shown, awarded on how completely and how well the work is done – not as all-or-nothing. There is no resubmission, so read this section before you start, not after.

What is assessed	Points
Analysis and reconciliation. Every disagreement between predicted and measured either explained or explicitly owned as unexplained, with the candidate causes ruled in or out.	20
The two matches. $VSWR \leq 2$ at 915 MHz by trimming and at f_2 by matching, both with traces shown – or, if a bar was missed, a quantitative diagnosis of why.	15
Pattern measurements. Co- and cross-polarized cuts in both planes at both frequencies, with the measured floor stated as a number and drawn on the plots.	15
Prediction before design. The NEC prediction of $Z_a(f_2)$ presented, and presented as what the match was designed against.	10
Gain by comparison. The substitution measurement, the reference antenna's own pattern, and the validity conditions addressed.	10
Report quality. All required figures and tables present, typed, legible, captioned, axes labeled, with the data files.	10
The balun. Built, and evidence in the report that it does something.	8
Instrument comparison. Both analyzers' traces of the final antenna, overlaid at the same reference plane, with the disagreement discussed.	7
Honest normalization. Every overlay's comparison basis declared in its caption.	5
Total	100

Two things are worth saying plainly about that table. **Analysis is the largest single row**, larger than either measurement row, because a campaign you cannot interpret is not a result. And a report in which everything agrees, nothing is questioned, and no measurement appears to have surprised its author will lose most of those twenty points however clean the data is.

Work that is missing entirely earns nothing for that row. Work that is present but thin – a pattern plotted without its floor, a match verified but not explained – earns part of it.

13 Checkpoints

These are not graded. They exist because the chamber is a shared resource and because a problem found at Lesson 11 is an afternoon, while the same problem found on 1 October is a failure.

By	You should have
Lesson 11	A trimmed dipole with a balun on it, $VSWR \leq 2$ at 915 MHz, shown to the instructor.
Lesson 14	f_2 declared, $Z_a(f_2)$ predicted in NEC, the L-network designed on paper.
Lesson 15	Matching network built and verified on the Copper Mountain. Chamber time booked.
Lesson 17	Chamber measurements complete, including the reference antenna.
2 October, 2359	Report and data files submitted.

Chamber time is scheduled in pairs. **You may share a chamber session and help each other take data. The data reduction, the analysis, and the report are individual.** Two reports containing the same sentences are two reports that are not your own work.

14 Authorized resources

Authorized without restriction: the course site and every lesson, lab, and practice set in it; your own notes; the course textbook; the NEC2 simulator; Python and its scientific libraries for reduction and plotting; the instrument manuals; and your instructor.

Discussion with classmates about method is encouraged. Sharing measured data, figures, or written text is not.

Generative AI use on this project is Level 3. If you use it, include a documentation statement, as the syllabus requires on any assignment – name the tool, what you used it for, and where it appears in your work.

The litmus test from the syllabus applies here as it does everywhere else: you must be able to explain every step of what you submit.

Questions about scope, equipment, or scheduling go to the instructor early. A question asked at Lesson 12 is free.